Appendix：A Comparative Explanation of Causal Inference Terminology in Medicine and End to End Autonomous Driving (AD) Models

Intervention

＊Medicine ; causal inference for treatment effects

Surgery, medication, therapeutic procedure

＊AD Models ; causal inference for action (task) selection

The intervention corresponds to the action selected by the autonomous driving model. (acceleration, braking and stopping, right or left steering)

Confounding

＊Medicine

Background differences distort treatment effect estimates

(severe patients receive aggressive therapy)

＊AD Models

Environmental factors influence both action selection and outcomes

(On narrow roads, night-time conditions combined with low-μ road surfaces increase both braking frequency and accident risk.)

Control Group: a set of observational instances used as a reference for causal comparison, typically consisting of units that did not receive the intervention under comparable conditions

＊Medicine

Patients who did not receive the treatment

＊AD Models

Unselected alternative actions (tasks) correspond to hypothetical scenarios—"what if it had not braked (breaking)?" or "what if it had gone straight instead of turning (steering)?"

These represent counterfactual outcomes, i.e., unobserved results under alternative actions (tasks) in the same situation

In contrast, the control group in the context of autonomous driving should be understood as empirically observed instances in which different actions (tasks) were taken under comparable conditions, providing real-world reference data for causal comparison.

In contrast, **Treatment Group**

＊Medicine

Patients who actually received the treatment

＊AD Models

The set of actions (tasks) the model actually selected

(executed a hard brake, turned right or left, stopped)

Covariates

＊Medicine
Background factors influencing treatment choice and outcomes
(age, comorbidities, severity, lifestyle, family medical history)
＊AD Models
Environmental and vehicle factors influencing action (task) choice and outcomes
(weather, night-time, congestion, pedestrian density, ODD conditions, sensor state)

Propensity Score
＊Medicine
Probability that a patient receives a treatment given covariates
＊AD Models
Probability that the model selects a specific action (task) given the observed situation
It is implicitly encoded within the model's internal decision-making structure.
In other words, it is implicitly associated with the model's internal purpose hierarchy and can be formally approximated from the model policy or decision function.

＊＊＊＊＊＊＊

A covariate is a baseline clinical factor that may influence treatment selection and/or patient outcomes. Among these, variables that causally affect both are confounders and must be adjusted for, as they open backdoor paths.
In the context of autonomous driving, the counterparts to covariates are baseline environmental and vehicle-state factors that may influence both the selection of intervention tasks (such as braking or steering) and accident risk.
Among these, factors that causally affect both function as confounders, as they open backdoor paths and must be statistically adjusted for.

A propensity score is the probability that a patient with given baseline covariates will receive a particular treatment.
1. More severe patients are more likely to receive aggressive treatments.
2. Younger patients are more likely to receive invasive therapies.
3. Older patients are more likely to receive conservative rather than surgical treatments.

In autonomous driving, a propensity score $PSe(x) = P(\text{intervention} \mid X = x)$ represents the probability that a vehicle will select a particular intervention action (task) given observed covariates such as weather, night-time conditions, traffic congestion, pedestrian density, ODD constraints, and sensor state.

＊＊＊＊＊＊＊

Prepared on March 21, 2026